Model Reference Adaptive Control (MRAC) Project

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1 Derivation of the Normalized MIT Rule with Disturbance Rejection

To handle the ambient temperature disturbance $T_a(t)$ effectively, the standard control law is augmented with a feedforward term. The goal is to derive the adaptation laws for the controller parameters so that the plant output tracks the reference model output perfectly.

1.1 System Definitions

Plant Model:

$$\dot{T} = -a_p T + b_p u + a_p T_a \tag{1}$$

Reference Model:

$$\dot{T}_m = -a_m T_m + b_m T_r \tag{2}$$

Control Law: We introduce a third parameter θ_3 to cancel the disturbance:

$$u(t) = \theta_1 T_r(t) - \theta_2 T(t) - \theta_3 T_a(t) \tag{3}$$

1.2 The Optimization Goal

We define the tracking error e and the cost function J (squared error) that we wish to minimize:

$$e(t) = T(t) - T_m(t) \tag{4}$$

$$J(\theta) = \frac{1}{2} e^2(t) \tag{5}$$

1.3 The MIT Rule (Gradient Descent)

The "MIT Rule" update law is:

$$\frac{d\theta}{dt} = -\gamma \frac{\partial J}{\partial \theta} = -\gamma \cdot e \cdot \frac{\partial e}{\partial \theta} \tag{6}$$

Since $e = T - T_m$ and T_m is independent of θ:

$$\dot{\theta} = -\gamma \cdot e \cdot \frac{\partial T}{\partial \theta} \tag{7}$$

1.4 Deriving Sensitivities ($\partial T/\partial \theta$)

Substituting the control law into the plant equation and applying the Laplace operator s:

$$T(s + a_p + b_p\theta_2) = b_p\theta_1 T_r - (b_p\theta_3 - a_p)T_a \tag{8}$$

Using the **Perfect Matching Approximation** (assuming $s + a_p + b_p\theta_2 \approx s + a_m$):

$$T \approx \frac{b_p\theta_1}{s + a_m}T_r - \frac{b_p\theta_3 - a_p}{s + a_m}T_a \tag{9}$$

We define the filtered signals (sensitivities) as follows:

- **For θ_1:** $\phi_1 = \frac{1}{s+a_m}T_r$
- **For θ_2:** $\phi_2 = \frac{1}{s+a_m}T$
- **For θ_3:** $\phi_3 = \frac{1}{s+a_m}T_a$

1.5 Final Normalized Control Laws

To ensure stability, we normalize by the squared norm of the sensitivity vector plus a small constant α:

$$\dot{\theta}_1 = -\frac{\gamma \cdot e \cdot \phi_1}{\alpha + \phi_1^2 + \phi_2^2 + \phi_3^2} \tag{10}$$

$$\dot{\theta}_2 = +\frac{\gamma \cdot e \cdot \phi_2}{\alpha + \phi_1^2 + \phi_2^2 + \phi_3^2} \tag{11}$$

$$\dot{\theta}_3 = +\frac{\gamma \cdot e \cdot \phi_3}{\alpha + \phi_1^2 + \phi_2^2 + \phi_3^2} \tag{12}$$

2 Calculated System Parameters

Based on student inputs $a = 1, b = 2$:

Table 1: System Parameter Values

Parameter	**Symbol**	**Value**	**Unit**
Plant Pole	a_p	0.01	s^{-1}
Plant Gain	b_p	0.000125	$°C/J$
Model Pole	a_m	0.02	s^{-1}
Model Gain	b_m	0.02	s^{-1}
Adaptation Gain	γ	40	-
Max Power	u_{max}	2500	W
Min Power	u_{min}	-1200	W

3 Calculation of Optimal Controller Parameters

To achieve perfect model matching, we equate the closed-loop plant dynamics to the reference model:

1. **Reference Input:** $\theta_1^* = b_m/b_p = 0.02/0.000125 = \mathbf{160}$
2. **System Dynamics:** $\theta_2^* = (a_m - a_p)/b_p = (0.02 - 0.01)/0.000125 = \mathbf{80}$
3. **Disturbance Rejection:** $\theta_3^* = a_p/b_p = 0.01/0.000125 = \mathbf{80}$

4 Anti-Windup Procedure

When the control signal $u(t)$ reaches the limits u_{max} or u_{min}, the adaptation is paused to prevent the parameters θ from growing unbounded.

Algorithm:

$$\dot{\theta}_{final} = \begin{cases} 0 & \text{if } u_{calc} > u_{max} \text{ or } u_{calc} < u_{min} \\ \dot{\theta}_{MIT} & \text{otherwise} \end{cases} \tag{13}$$

5 Simulation Results and Discussion

This section presents the performance of the MRAC controller under various operating conditions. The simulations were performed to verify tracking accuracy and disturbance rejection.

5.1 Set Parameters and Initialization

The following figures show the constants and the initialization parameters used in the MATLAB workspace before running the simulation.

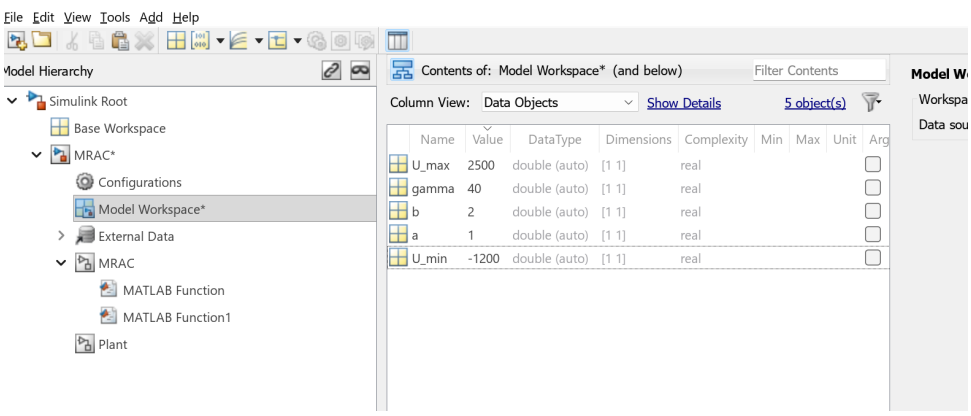

Figure 1: System Physical Constants

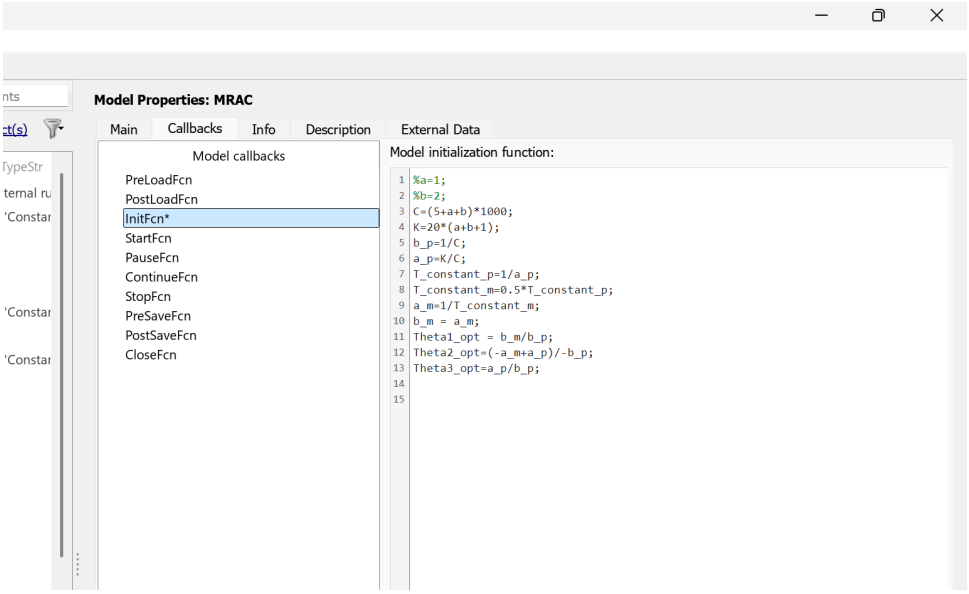

Figure 2: Initialization of Controller Parameters

5.2 Simulink Models

The implementation was carried out in Simulink. The plant model, reference model, and the MRAC adaptation mechanism are shown below.

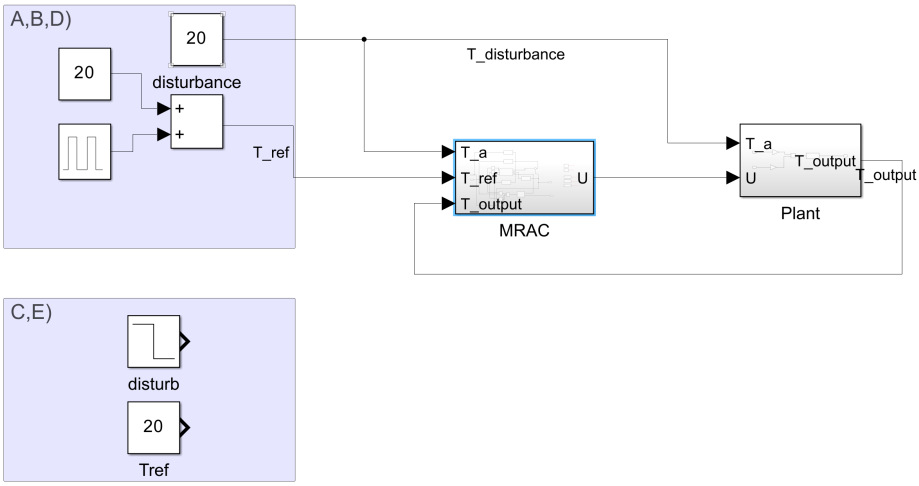

Figure 3: Main Simulink Block Diagram

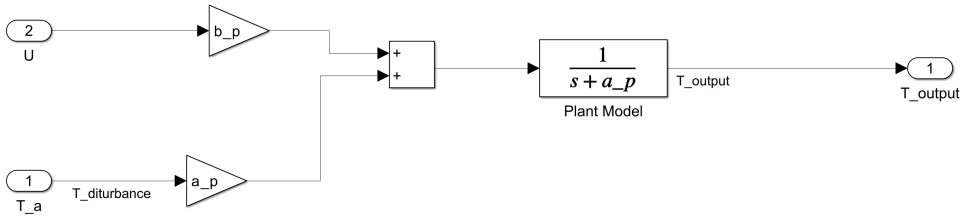

Figure 4: Internal Model of the Thermal Plant

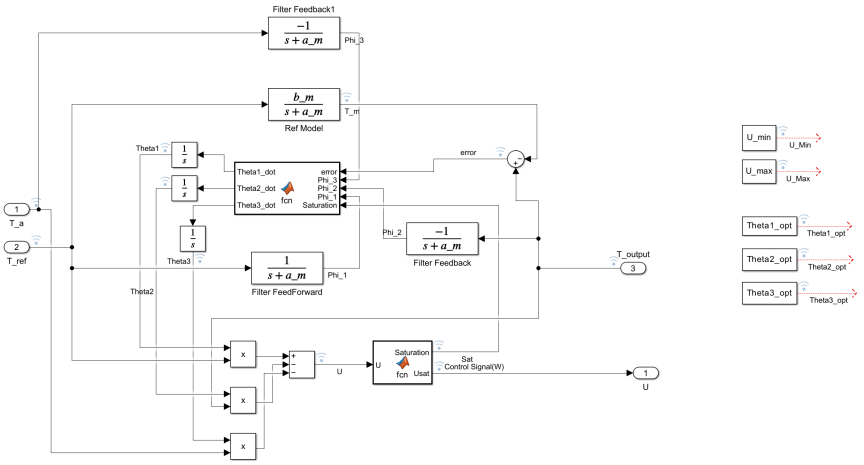

Figure 5: Subsystem of the MRAC Adaptation Law

5.3 Settling of Control Law Parameters

These plots show how the adaptive gains θ_1, θ_2, and θ_3 converge over time to their optimal values.

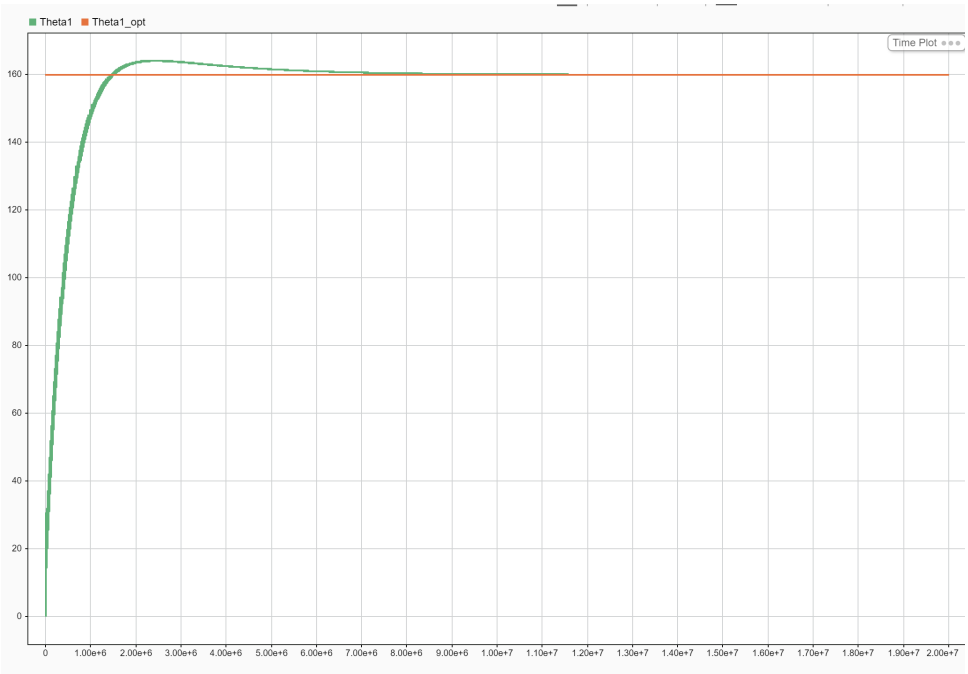

Figure 6: Adaptation of Feedforward Gain θ_1

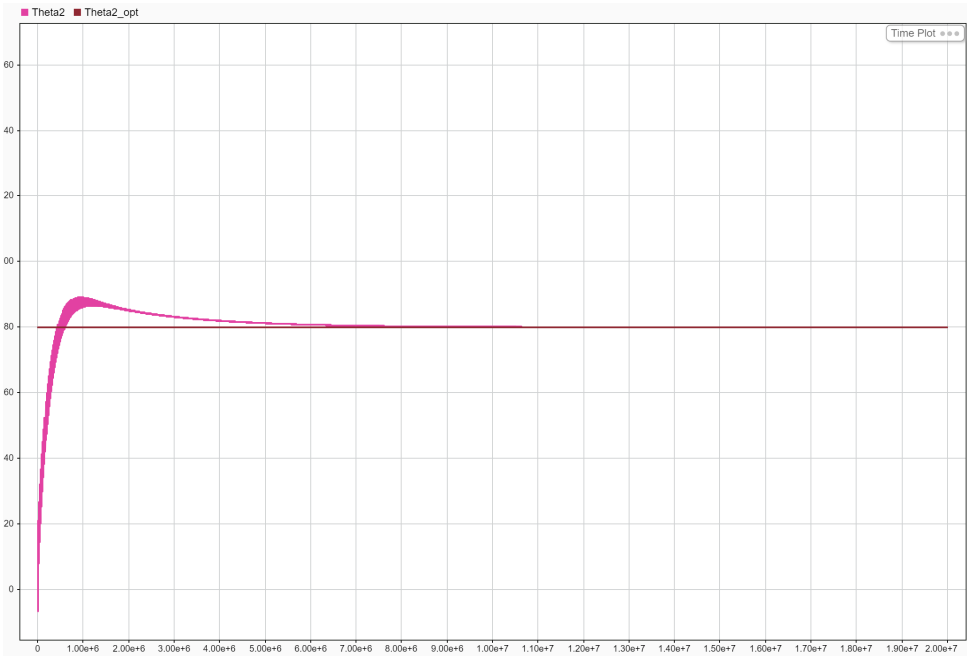

Figure 7: Adaptation of Feedback Gain θ_2

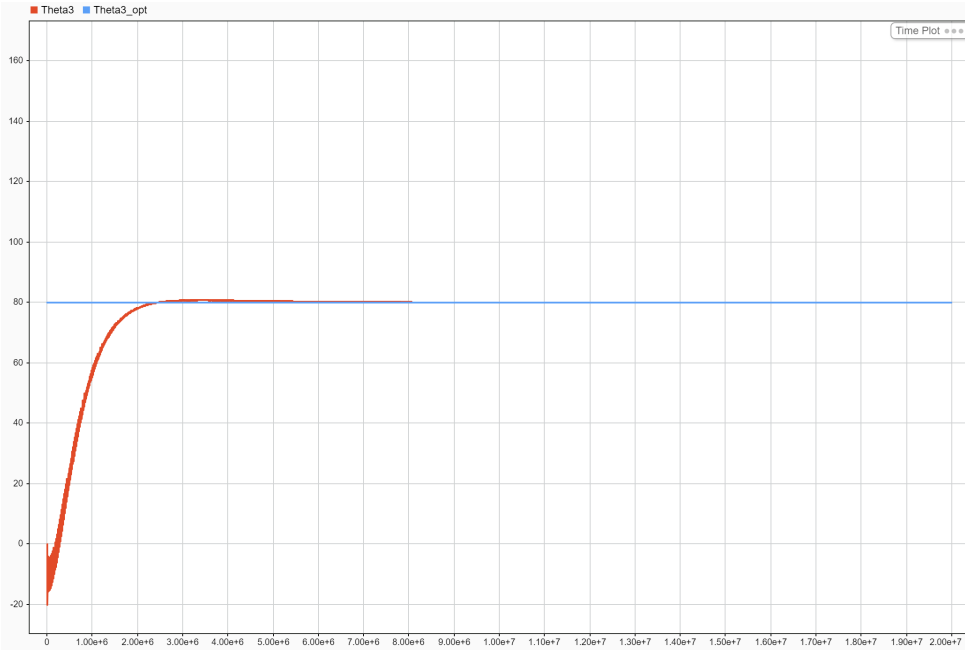

Figure 8: Adaptation of Disturbance Rejection Gain θ_3

5.4 Control Signal

The control signal $u(t)$ represents the power sent to the heater. Note how the signal behaves when saturation limits are reached.

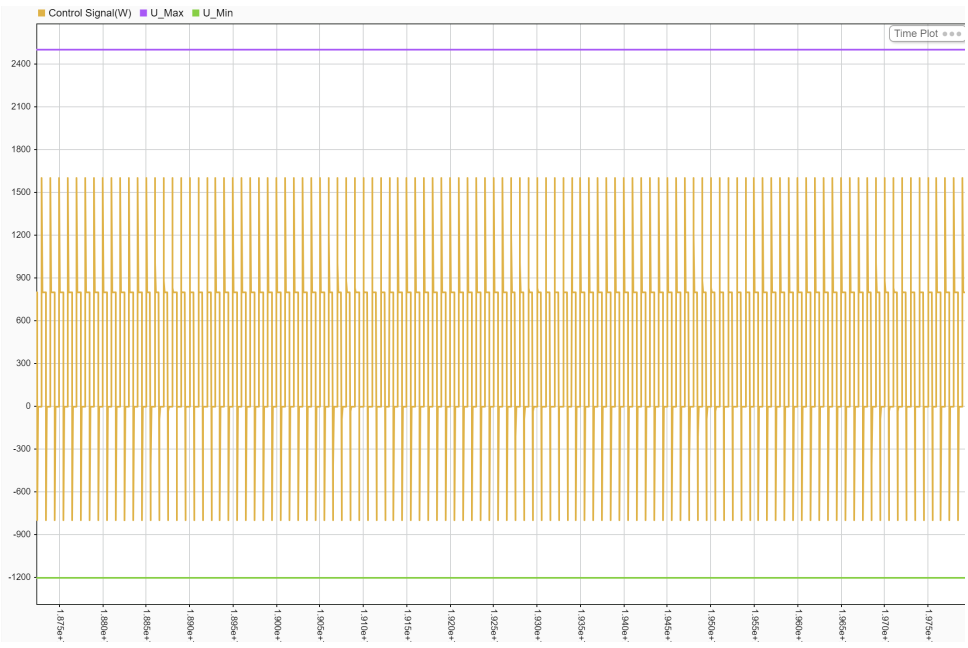

Figure 9: Control Signal $u(t)$ over time

5.5 Output Signal

The primary goal is for the plant temperature T to follow the reference model T_m.

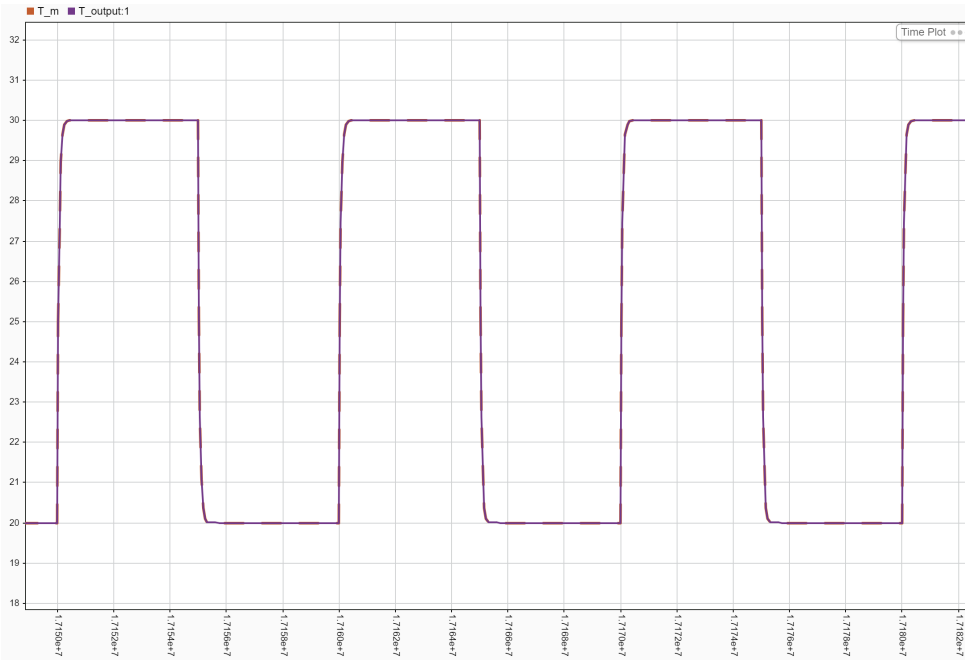

Figure 10: Comparison of Plant Output (T) and Reference Model (T_m)

5.6 Error

The tracking error $e(t) = T(t) - T_m(t)$ should converge toward zero as the parameters adapt.

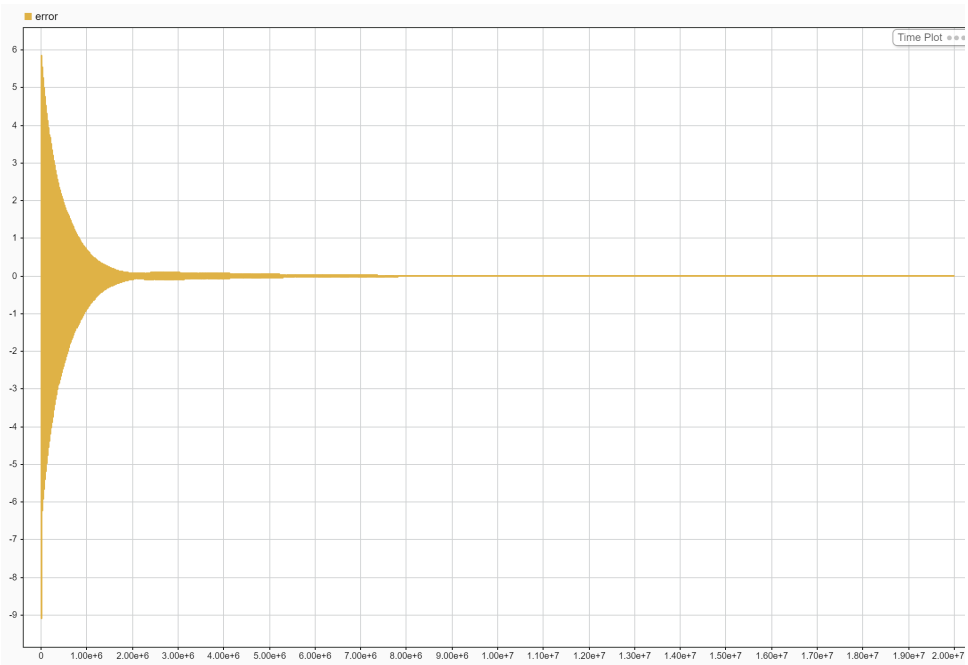

Figure 11: Tracking Error Signal $e(t)$

6 Test Cases

This section presents the performance of the MRAC controller under various operating conditions. The experiments verify the system's tracking accuracy, disturbance rejection, and robustness to noise and saturation.

6.1 Test Case A: Adaptation Coefficients Tuning

Objective: Experimentally determine the coefficients responsible for the adaptation rate without control signal limitations.

- **Setup:** Adaptation gain $\gamma = 40$, no saturation limits.
- **Observation:** The adaptive parameters θ_1, θ_2, and θ_3 converge to their theoretical optimal values, and the tracking error approaches zero.

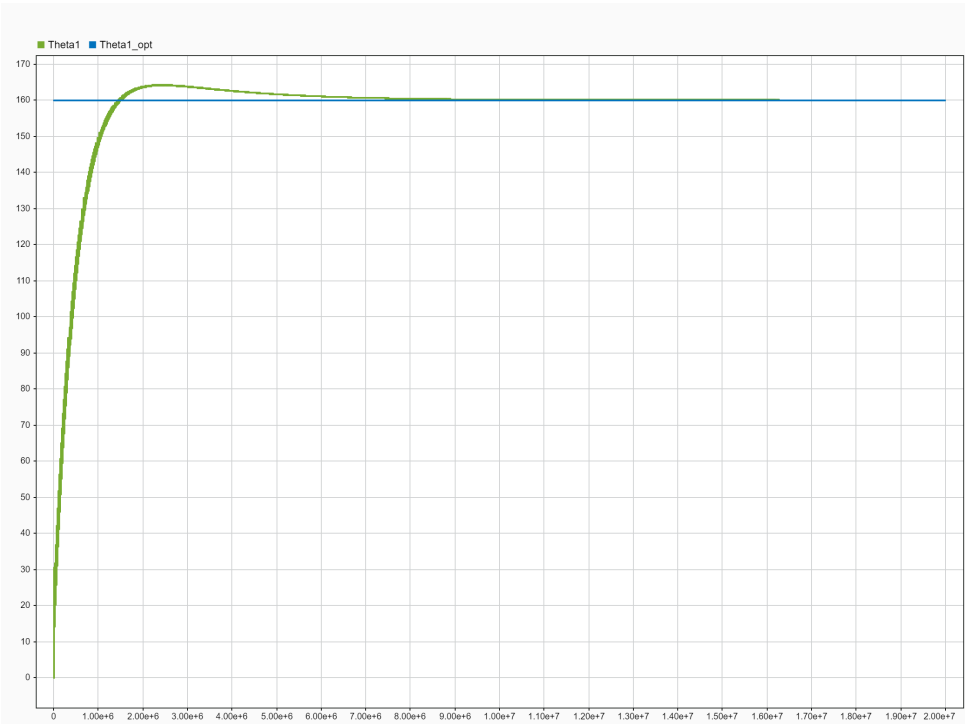

Figure 12: Case A: Convergence of Feedforward Gain θ_1.

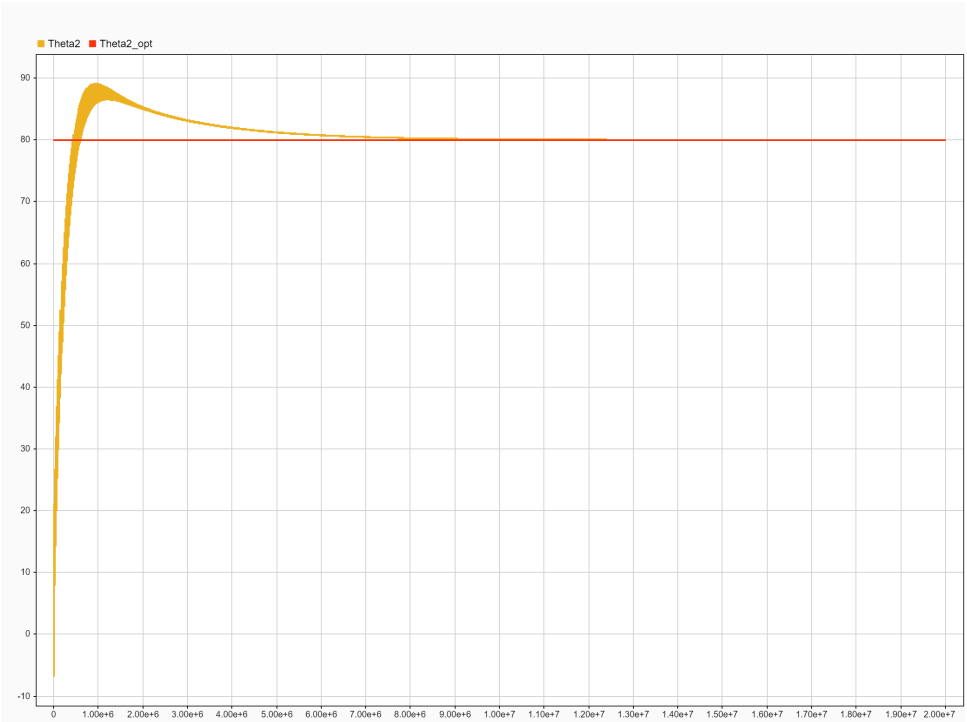

Figure 13: Case A: Convergence of Feedback Gain θ_2.

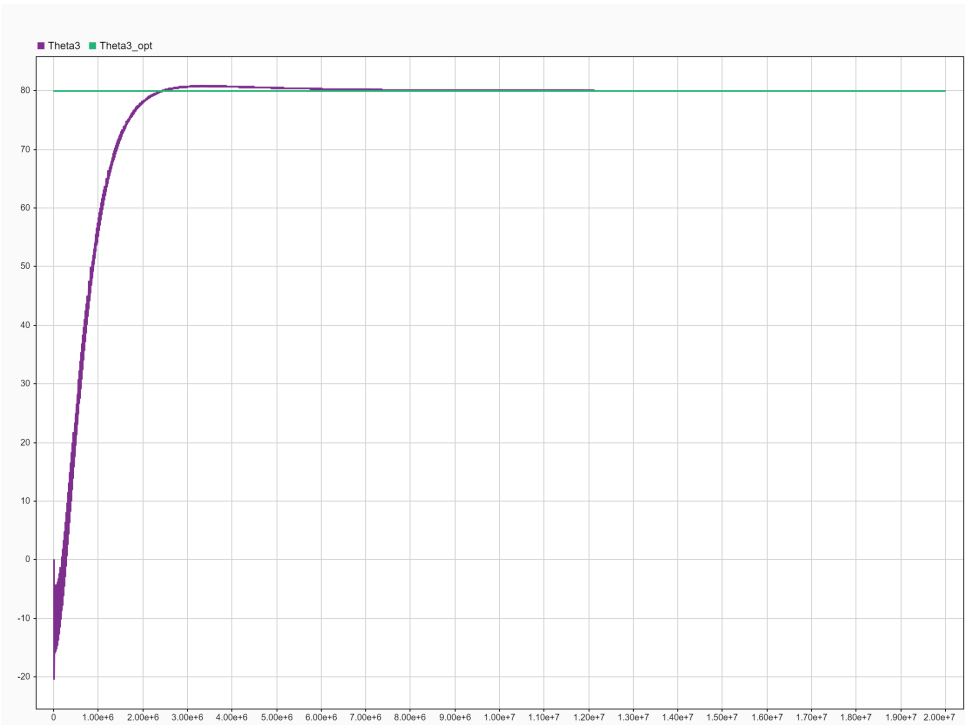

Figure 14: Case A: Convergence of Disturbance Rejection Gain θ_3.

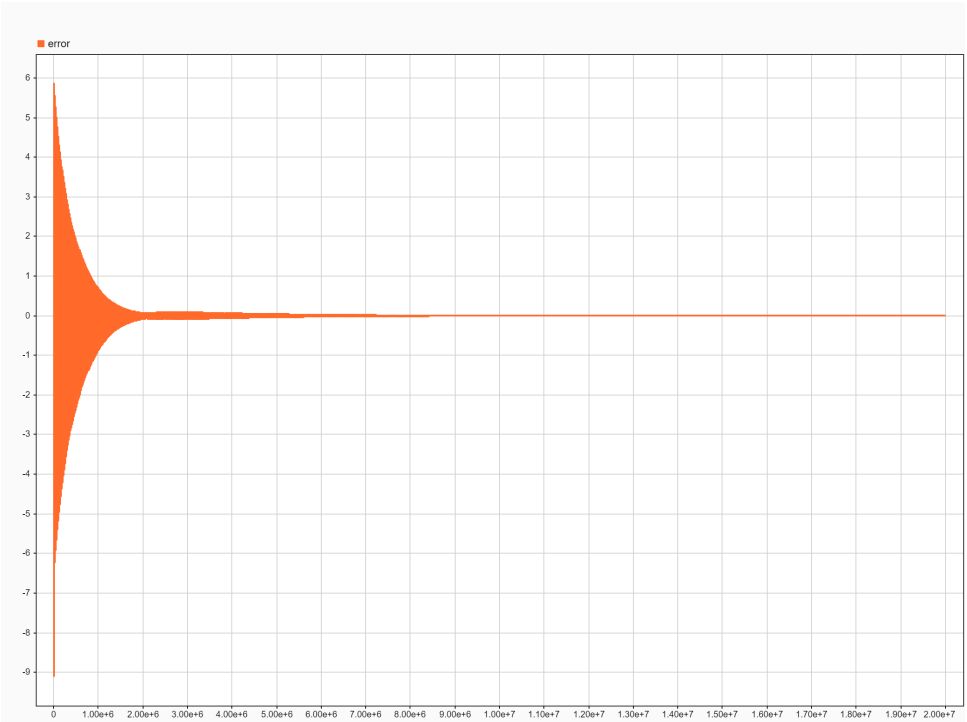

Figure 15: Case A: Evolution of the tracking error $e(t)$ over time.

6.2 Test Case B: Reference Step Response (No Limits)

Objective: Determine the closed-loop response to a step change in reference temperature from $20°C$ to $30°C$ with constant ambient temperature $T_a = 20°C$, ignoring control limits.

- **Observation:** The plant output $T(t)$ perfectly tracks the reference model $T_m(t)$, demonstrating ideal model matching.

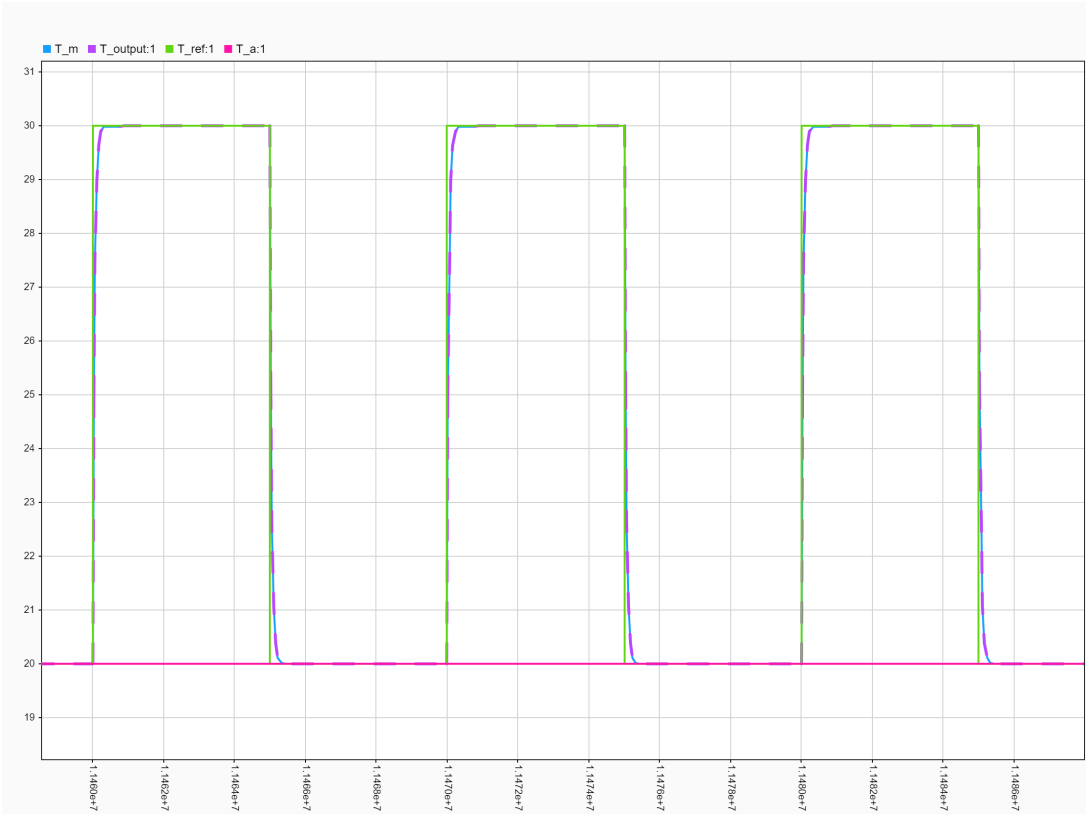

Figure 16: Case B: Temperature tracking (T, T_m, T_r) and Ambient Temperature (T_a).

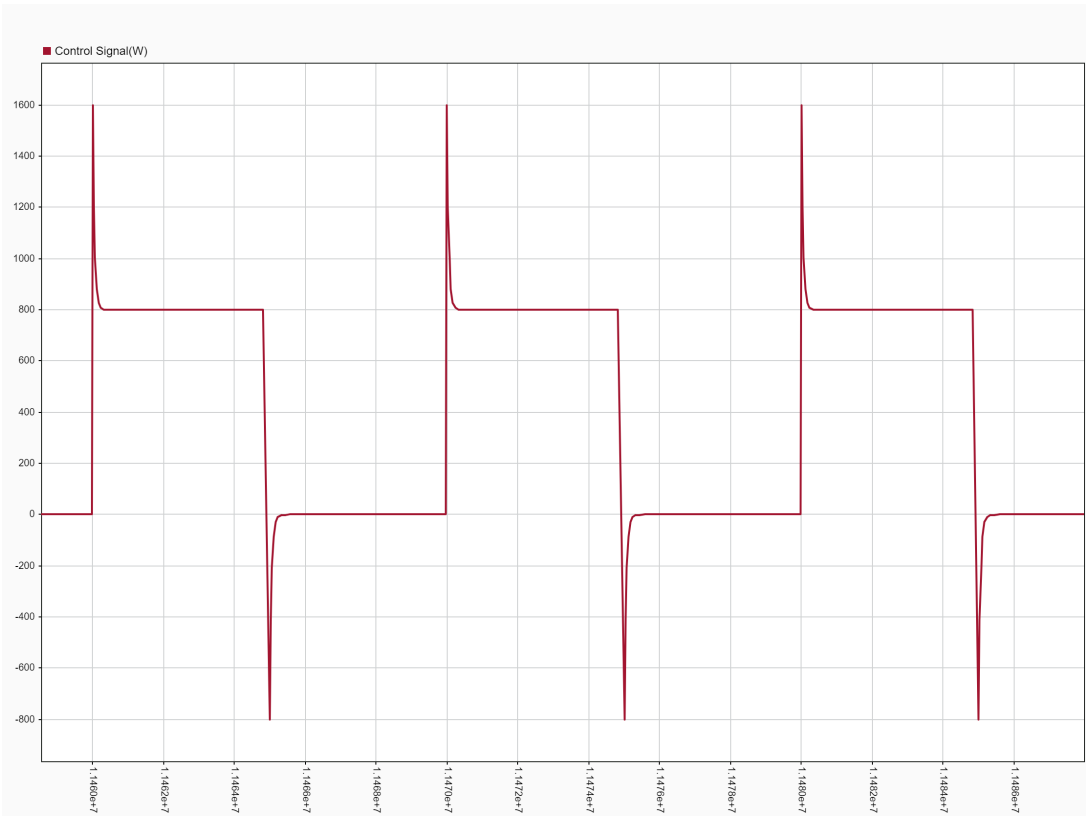

Figure 17: Case B: Control Signal $u(t)$ required for the step response (Unconstrained).

6.3 Test Case C: Disturbance Rejection (No Limits)

Objective: Determine the response to a step change in ambient temperature $T_a(t)$ from $20°C$ to $15°C$ with constant reference $T_r = 20°C$.

- **Observation:** The controller effectively increases the heating power to compensate for the drop in ambient temperature, maintaining the object temperature at the setpoint.

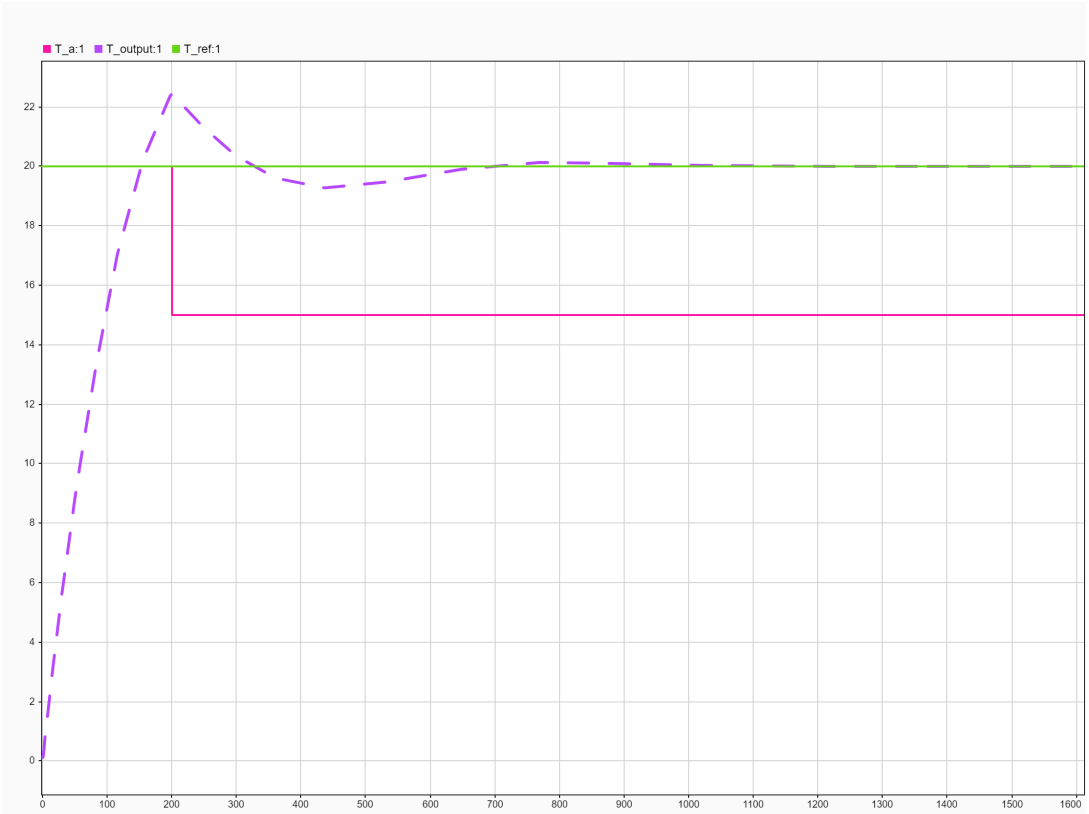

Figure 18: Case C: Effect of ambient temperature drop on plant output.

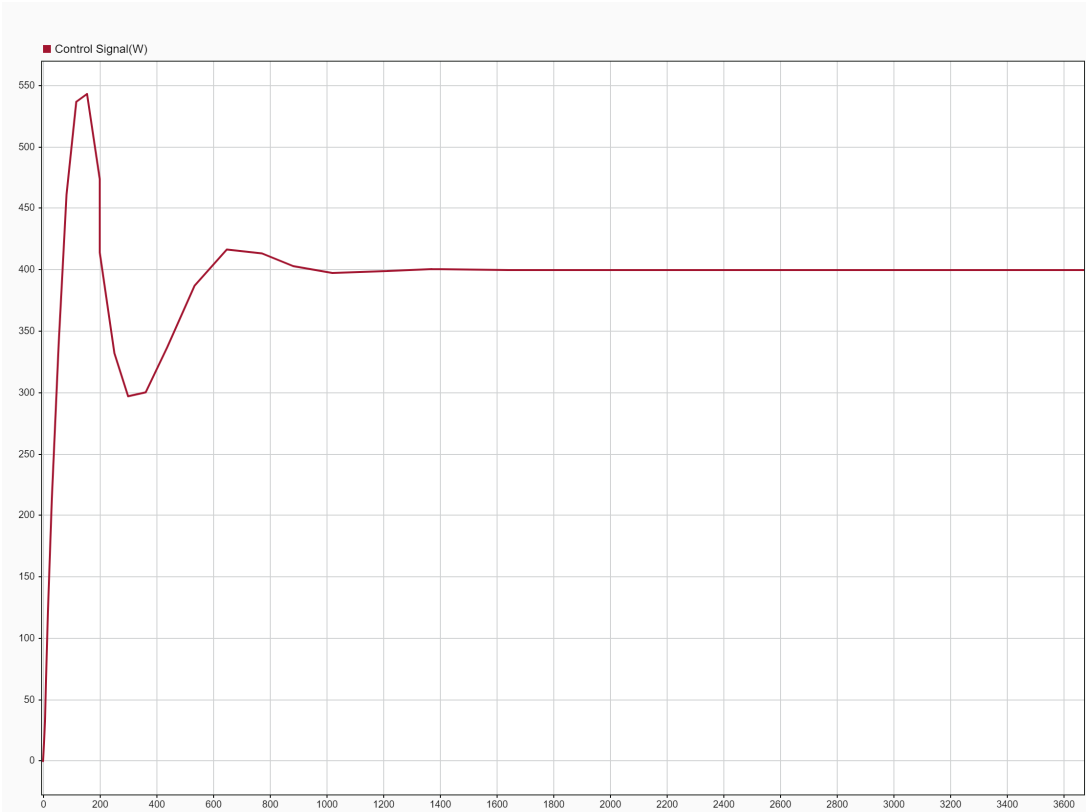

Figure 19: Case C: Control effort increasing to reject the cold disturbance.

6.4 Test Case D: Reference Step Response (With Limits)

Objective: Repeat the reference step change ($20°C \rightarrow 30°C$) while considering control signal constraints ($u_{max} = 900W, u_{min} = -600W$).

- **Observation:** The control signal saturates at u_{max}. The anti-windup mechanism pauses adaptation during saturation, causing a temporary lag in tracking compared to the unconstrained case.

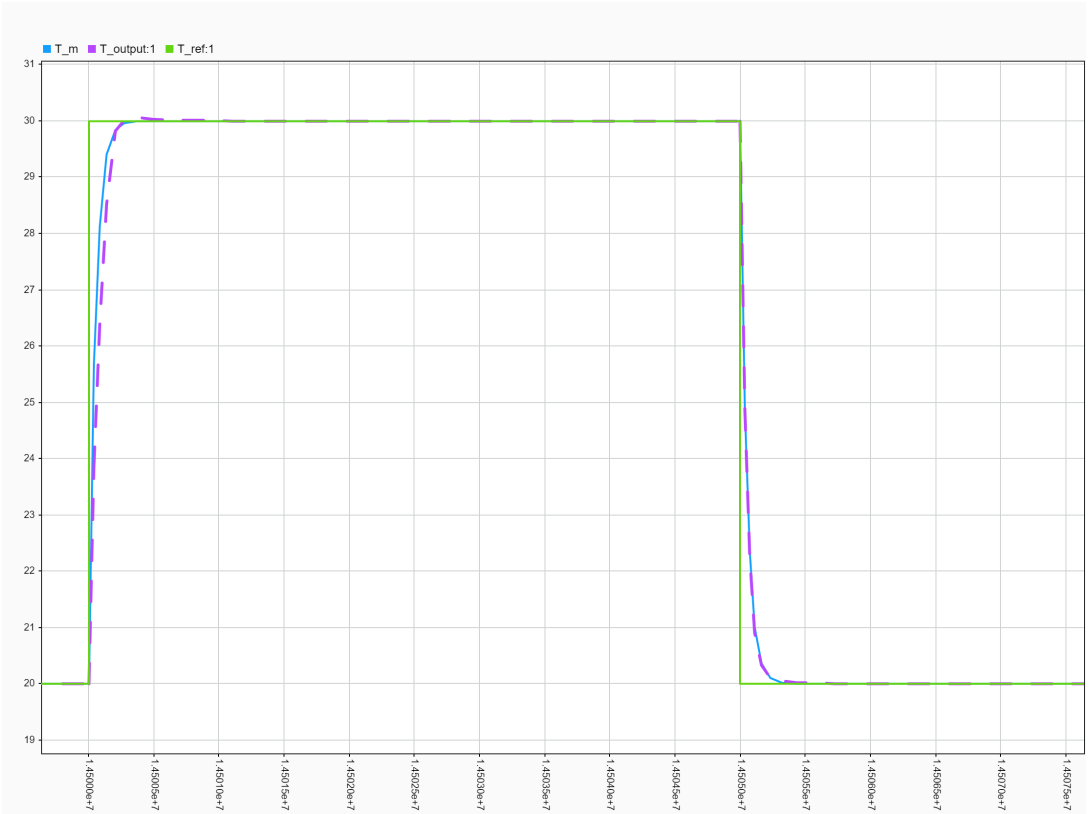

Figure 20: Case D: Temperature tracking performance under actuator saturation.

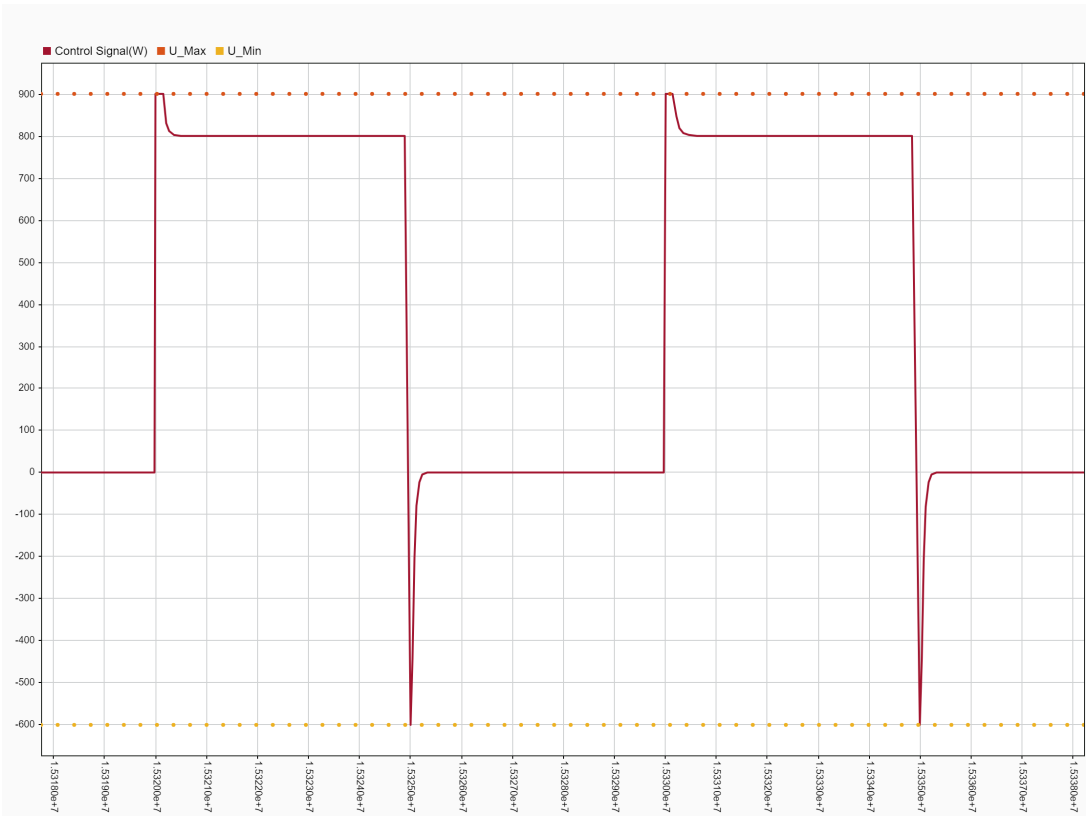

Figure 21: Case D: Control signal hitting the saturation limit u_{max}.

6.5 Test Case E: Disturbance Rejection (With Limits)

Objective: Repeat the disturbance step change ($T_a : 20°C \rightarrow 15°C$) while considering control signal constraints.

- **Observation:** The controller adjusts power to reject the disturbance. The signal is verified against u_{max} and u_{min} to ensure safe operation.

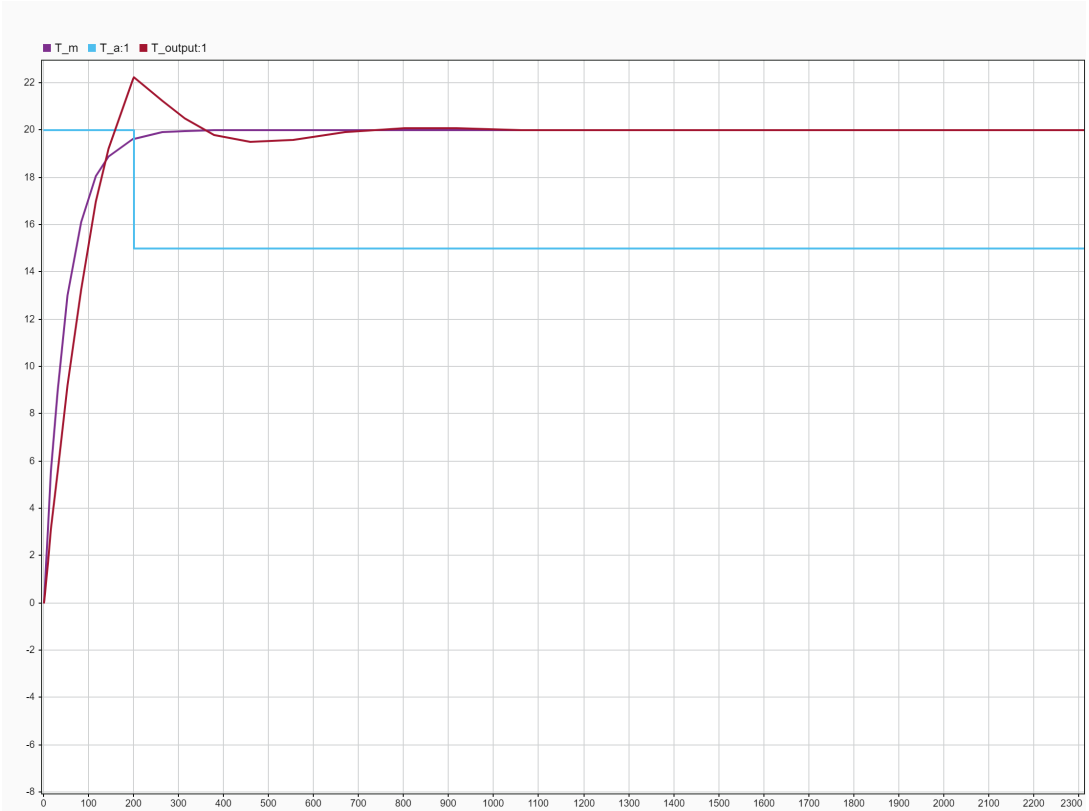

Figure 22: Case E: Disturbance rejection performance with limited power.

Figure 23: Case E: Control signal relative to saturation limits.

6.6 Test Case F: Robustness to Measurement Noise

Objective: Reference step change ($20°C \rightarrow 30°C$) with measurement noise (SD $= 0.5°C$) added to the feedback path.

- **Observation:** The plot displays both the noisy measured signal fed to the controller and the actual physical plant output. Despite the noise, the MRAC system maintains stability.

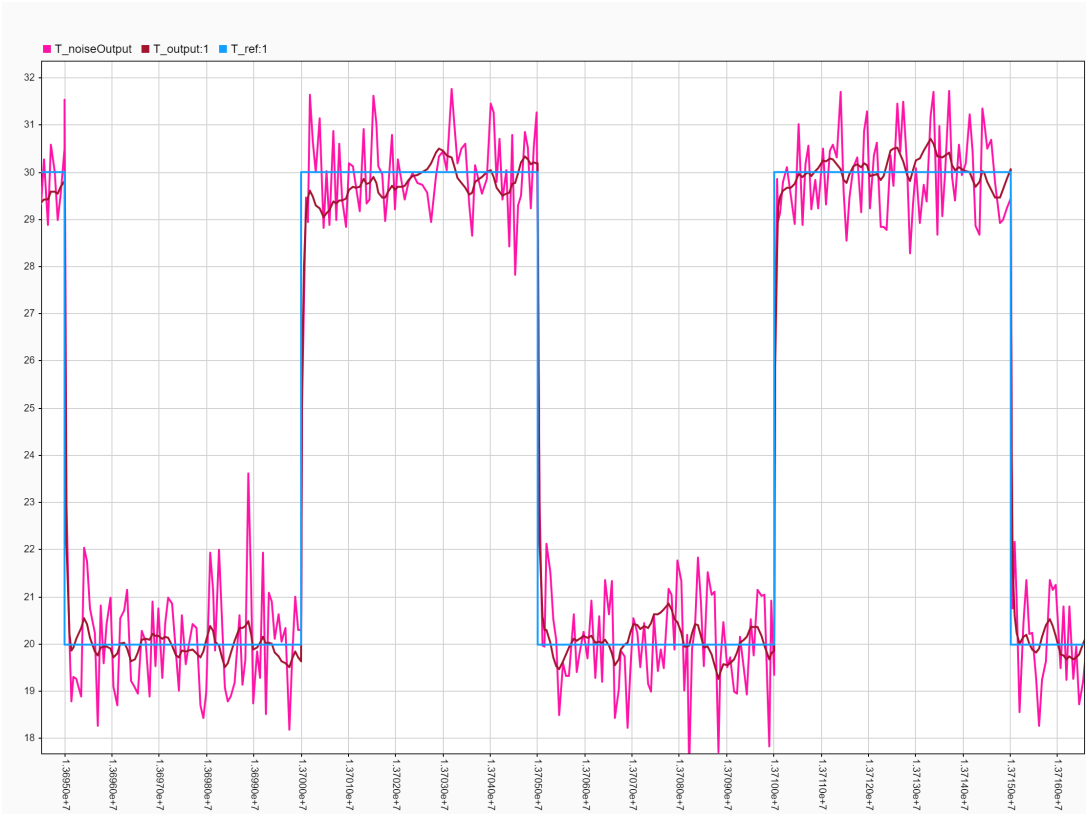

Figure 24: Case F: Comparison of noisy measured temperature vs. actual plant output.

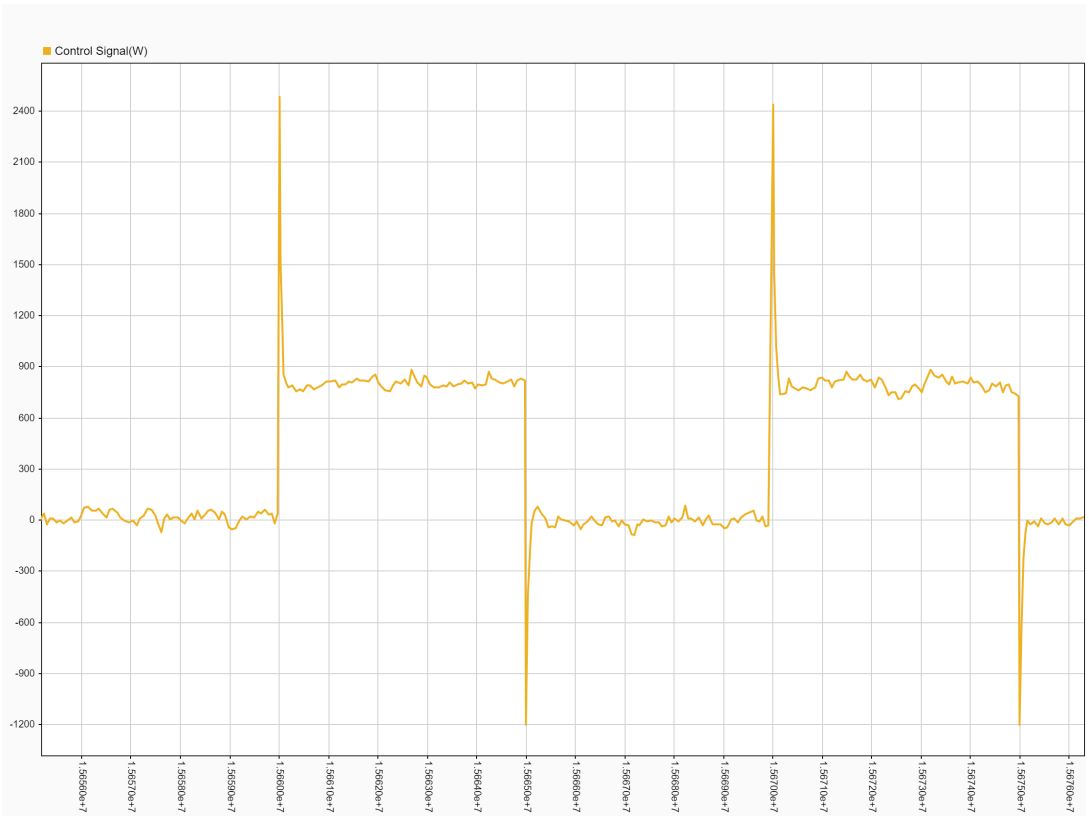

Figure 25: Case F: Control signal fluctuations due to measurement noise.